

DBMS Week - 2

Question1: Create Table

```
mysql> create table student_marks1 (  
-> roll_no int primary key,  
-> name varchar(50),  
-> subject varchar(50),  
-> marks decimal(5,2)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> show tables;
```

```
+-----+  
| Tables_in_task2 |  
+-----+  
| student_marks1 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> desc student_marks1;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| roll_no | int | NO | PRI | NULL | |  
| name | varchar(50) | YES | | NULL | |  
| subject | varchar(50) | YES | | NULL | |  
| marks | decimal(5,2) | YES | | NULL | |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

Question2. Insert Data

```
mysql> insert into student_marks1 (roll_no, name, subject, marks) values  
-> (1, 'ravi', 'math', 85.50),  
-> (2, 'sita', 'math', 92.75),  
-> (3, 'anil', 'math', 78.40),  
-> (4, 'priya', 'math', 88.90),  
-> (5, 'vijay', 'math', 80.25),  
-> (6, 'subbusir', 'aws', 98.50),  
-> (7, 'dwaraka', 'dbms', 95.75),  
-> (8, 'ranjani', 'english', 97.40),  
-> (9, 'kavithamam', 'aws1', 99.90),  
-> (10, 'seetha', 'azure', 82.25);
```

Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

Question3. Simple Select

```
mysql> select * from student_marks1;
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
3	anil	math	78.40
4	priya	math	88.90
5	vijay	math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.00 sec)

mysql>
```

Question4. Aggregate Functions

```
mysql> select count(*) as total_students from student_marks1;
```

total_students
10

```
1 row in set (0.01 sec)

mysql> select sum(marks) from student_marks1;
```

sum(marks)
899.60

```
1 row in set (0.01 sec)

mysql> select avg(marks) from student_marks1;
```

avg(marks)
89.960000

```
1 row in set (0.00 sec)

mysql> select max(marks) from student_marks1;
```

max(marks)
99.90

```
1 row in set (0.01 sec)

mysql> select min(marks) from student_marks1;
```

min(marks)
78.40

```
1 row in set (0.00 sec)

mysql> |
```

5. Filtering with WHERE

Question

```
mysql> select * from student_marks1
-> where marks > 85;
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
4	priya	math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

7 rows in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks >= 90;
```

roll_no	name	subject	marks
2	sita	math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

5 rows in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks < 80;
```

roll_no	name	subject	marks
3	anil	math	78.40

1 row in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks between 80 and 90;
```

roll_no	name	subject	marks
1	ravi	math	85.50
4	priya	math	88.90
5	vijay	math	80.25
10	seetha	azure	82.25

4 rows in set (0.01 sec)

```
mysql> select * from student_marks1
-> where name like 'p%';
```

roll_no	name	subject	marks
4	priya	math	88.90

1 row in set (0.00 sec)

```
mysql> select * from student_marks1
-> where name in ('ravi', 'sita', 'vijay');
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
5	vijay	math	80.25

3 rows in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks > 85
-> and (subject = 'math' or name like 'p%');
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
4	priya	math	88.90

3 rows in set (0.00 sec)

```
mysql> |
```

6. Update Queries

```
mysql> update student_marks1
    -> set marks = 90.00
    -> where roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from student_marks1
    -> where roll_no = 3;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      3 | anil | math    | 90.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql> update student_marks1
    -> set subject = 'remedial math'
    -> where marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql> select * from student_marks1
    -> where marks < 80;
Empty set (0.00 sec)
```

```
mysql> |
```

Question7. Delete Queries

```
mysql> delete from student_marks1
    -> where roll_no = 5;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from student_marks1;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      1 | ravi | math    | 85.50 |
|      2 | sita | math    | 92.75 |
|      3 | anil | math    | 90.00 |
|      4 | priya | math    | 88.90 |
|      6 | subbusir | aws    | 98.50 |
|      7 | dwaraka | dbms    | 95.75 |
|      8 | ranjani | english | 97.40 |
|      9 | kavithamam | aws1    | 99.90 |
|     10 | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

```
mysql> delete from student_marks1
    -> where marks < 75;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from student_marks1;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      1 | ravi | math    | 85.50 |
|      2 | sita | math    | 92.75 |
|      3 | anil | math    | 90.00 |
|      4 | priya | math    | 88.90 |
|      6 | subbusir | aws    | 98.50 |
|      7 | dwaraka | dbms    | 95.75 |
|      8 | ranjani | english | 97.40 |
|      9 | kavithamam | aws1    | 99.90 |
|     10 | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

Question 8: sorting

```
mysql> select * from student_marks1  
-> order by marks asc;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	ravi	math	85.50
4	priya	math	88.90
3	anil	math	90.00
2	sita	math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

9 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> order by marks desc;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	sita	math	92.75
3	anil	math	90.00
4	priya	math	88.90
1	ravi	math	85.50
10	seetha	azure	82.25

```
mysql> select * from student_marks1  
-> order by name asc;
```

roll_no	name	subject	marks
3	anil	math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	priya	math	88.90
8	ranjani	english	97.40
1	ravi	math	85.50
10	seetha	azure	82.25
2	sita	math	92.75
6	subbusir	aws	98.50

9 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> order by name desc;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	sita	math	92.75
10	seetha	azure	82.25
1	ravi	math	85.50
8	ranjani	english	97.40
4	priya	math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	anil	math	90.00

9 rows in set (0.00 sec)

```
mysql> |
```

Question 9: Group

```
mysql> select subject, sum(marks) as total_marks  
-> from student_marks1  
-> group by subject;
```

subject	total_marks
math	357.15
aws	98.50
dbms	95.75
english	97.40
aws1	99.90
azure	82.25

6 rows in set (0.02 sec)

```
mysql> select subject, avg(marks) as avg_marks  
-> from student_marks1  
-> group by subject;
```

subject	avg_marks
math	89.287500
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000
azure	82.250000

6 rows in set (0.00 sec)

```
mysql> select subject, count(*) as num_students  
-> from student_marks1  
-> group by subject;
```

subject	num_students
math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

6 rows in set (0.00 sec)

```
mysql> |
```

Question10. Having

```
mysql> select subject, avg(marks) as avg_marks  
-> from student_marks1  
-> group by subject  
-> having avg(marks) > 90;
```

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

4 rows in set (0.00 sec)

```
mysql> select subject, count(*) as num_students  
-> from student_marks1  
-> group by subject  
-> having count(*) > 1;
```

subject	num_students
math	4

1 row in set (0.00 sec)

```
mysql> |
```